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COURSE CODE: ECA0910
COURSE NAME: DIGITAL SIGNAL PROCESSING
PRACTICAL LAB RESULTS

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Specifications
```

```
Fs = 2000;      % Sampling frequency (Hz)
```

```
Rp = 1;         % Passband ripple (dB)
```

```
Rs = 40;        % Stopband attenuation (dB)
```

```
Wp = [300 600]/(Fs/2); % Normalized passband
```

```
Ws = [200 700]/(Fs/2); % Normalized stopband
```

```
%% Butterworth Bandpass Filter
```

```
[n1, W1] = buttord(Wp, Ws, Rp, Rs);
```

```
[b1, a1] = butter(n1, W1, 'bandpass');
```

```
%% Chebyshev Type-I Bandpass Filter
```

```
[n2, W2] = cheb1ord(Wp, Ws, Rp, Rs);
```

```
[b2, a2] = cheby1(n2, Rp, W2, 'bandpass');
```

```
%% Display Filter Orders
```

```
fprintf('Butterworth Filter Order = %d\n', n1);  
fprintf('Chebyshev Type-I Filter Order = %d\n', n2);
```

```
%% Magnitude Response - Butterworth
```

```
figure;  
freqz(b1, a1, 1024, Fs);  
title('Butterworth Bandpass Filter');
```

```
%% Magnitude Response - Chebyshev Type-I
```

```
figure;  
freqz(b2, a2, 1024, Fs);  
title('Chebyshev Type-I Bandpass Filter');
```

```
%% Group Delay Comparison
```

```
figure;  
[gd1, f1] = grpdelay(b1, a1, 1024, Fs);  
[gd2, f2] = grpdelay(b2, a2, 1024, Fs);
```

```
plot(f1, gd1, 'LineWidth', 1.2);
```

```
hold on;
```

```
plot(f2, gd2, 'LineWidth', 1.2);
```

```
grid on;
```

```
xlabel('Frequency (Hz)');
```

```
ylabel('Group Delay (samples)');
```

```
title('Group Delay Comparison');
```

```
legend('Butterworth', 'Chebyshev Type-I');
```

```
%% Pole-Zero Plot
```

```
figure;
```

```
subplot(1,2,1);
```

```
zplane(b1, a1);
```

```
title('Butterworth Pole-Zero Plot');
```

```
subplot(1,2,2);
```

```
zplane(b2, a2);
```

```
title('Chebyshev Type-I Pole-Zero Plot');
```

```
%% Impulse Response
```

```
figure;
```

```
subplot(2,1,1);
```

```
impz(b1, a1);
```

```
title('Butterworth Impulse Response');
```

```
grid on;
```

```
subplot(2,1,2);
```

```
impz(b2, a2);
```

```
title('Chebyshev Type-I Impulse Response');
```

```
grid on;
```

```
%% Pole Magnitude
```

```
figure;
```

```
subplot(1,2,1);
```

```

stem(abs(roots(a1)), 'filled');

title('Butterworth Pole Magnitude');

xlabel('Pole Number');

ylabel('| Pole |');

grid on;

subplot(1,2,2);

stem(abs(roots(a2)), 'filled');

title('Chebyshev Type-I Pole Magnitude');

xlabel('Pole Number');

ylabel('| Pole |');

grid on;

```

OUTPUT:





